



Technical Service Bulletin: TSB250061

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New Accessory Drive Support: Excessive Engine Noise

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

- QSK23 CM500
- QSK23 CM2250 K109
- QSK23 CM2350 K127

Issue Summary

Symptom

- Excessive noise from front side of engine
- Seized and overheated accessory drive
- Worn or malfunctioning accessory drive bearings

Root Cause(s):

- Insufficient lubrication
- Excessive load applied to the accessory drive support
- Improper belt tension



Figure 1, Disassembled Accessory Drive Support, Bearing Malfunction / Shaft and Housing Damage.

Verification

Seized or malfunctioning accessory drive bearings resulting in damaged or malfunctioning adaptor assemblies. See Figure 1.

Resolution

- A new increased load bearing capacity accessory drive support with a rear cylindrical roller bearing has been released.
- Both front and rear bearings of accessory drive are now updated to cylindrical roller type.
- New accessory drive is equipped with direct feed external lubricating oil line.

Service Instructions

1. Install new accessory drive support. See corresponding Service Manual. Reference Procedure 009-009 in Section 9.
2. Install male union (see Figure 2, Item 4) in cylinder block after removing plug . Torque to 20 N•m [177 in-lb].
3. Loosely install elbow male union (see Figure 2, Item 2) to accessory drive external inlet port.
4. Loosely install hose (see Figure 2, Item 3). Verify appropriate angle of elbow is achieved as to avoid interference (fouling) and hand tighten.
5. Check for appropriate clearance to avoid interference / fouling of line with cylinder block, brackets, mounts, OEM frame.
6. Torque elbow (see Figure 2, Item 2) to 25 N•m [221 in-lb]. Torque hose fittings (see Figure 2, Item 3) to 16 N•m [142 in-lb].
7. When installing belts of manually tensioned accessories, see equipment manufacturer Service Information for proper belt tension. A general belt tension recommendation of 636 N [

143 lbf] is suggested.

8. Operate engine and check for leaks.

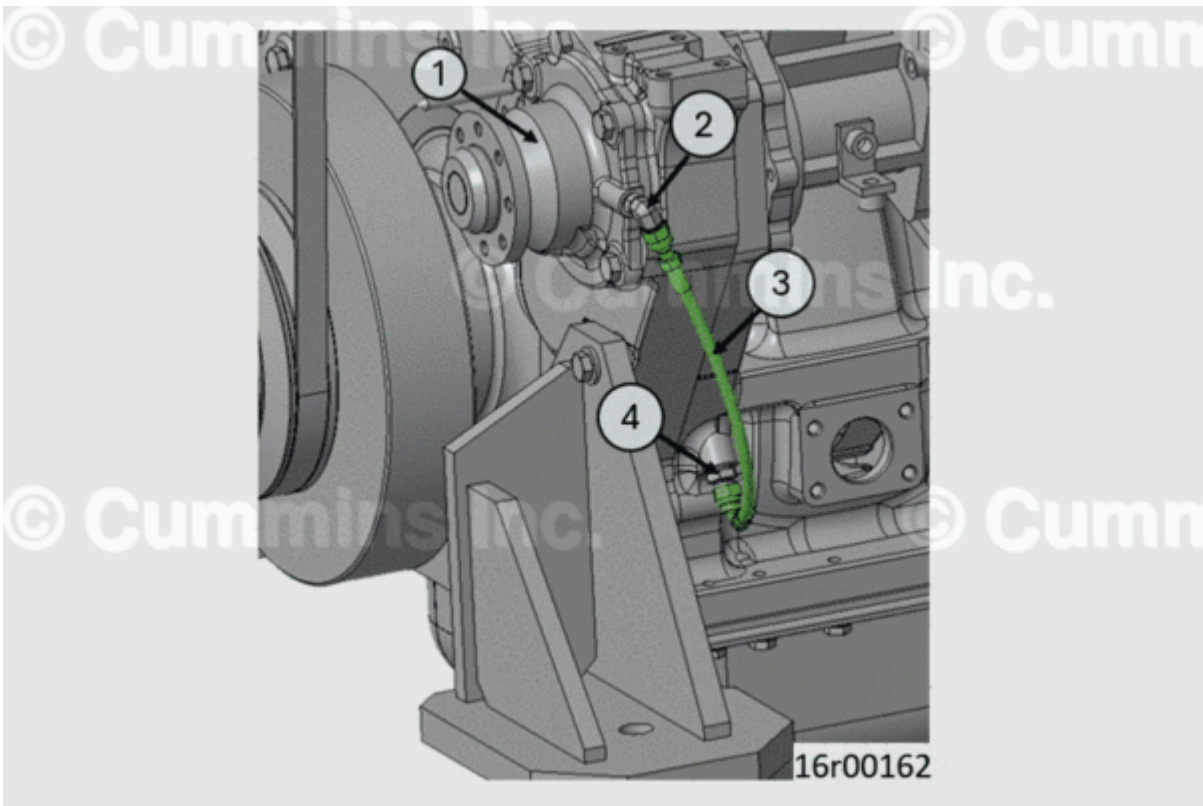


Figure 2, Assembly with Part Numbers Identified.

Table 1, Figure 2: Assembly with Part Numbers Identified.		
Figure 2, Item Number	Part Number	Name
1	6555990	Support, Accessory Drive
2	3089581	Elbow, Male Union
3	6499057	Hose, Flexible
4	3682300	Union, Male

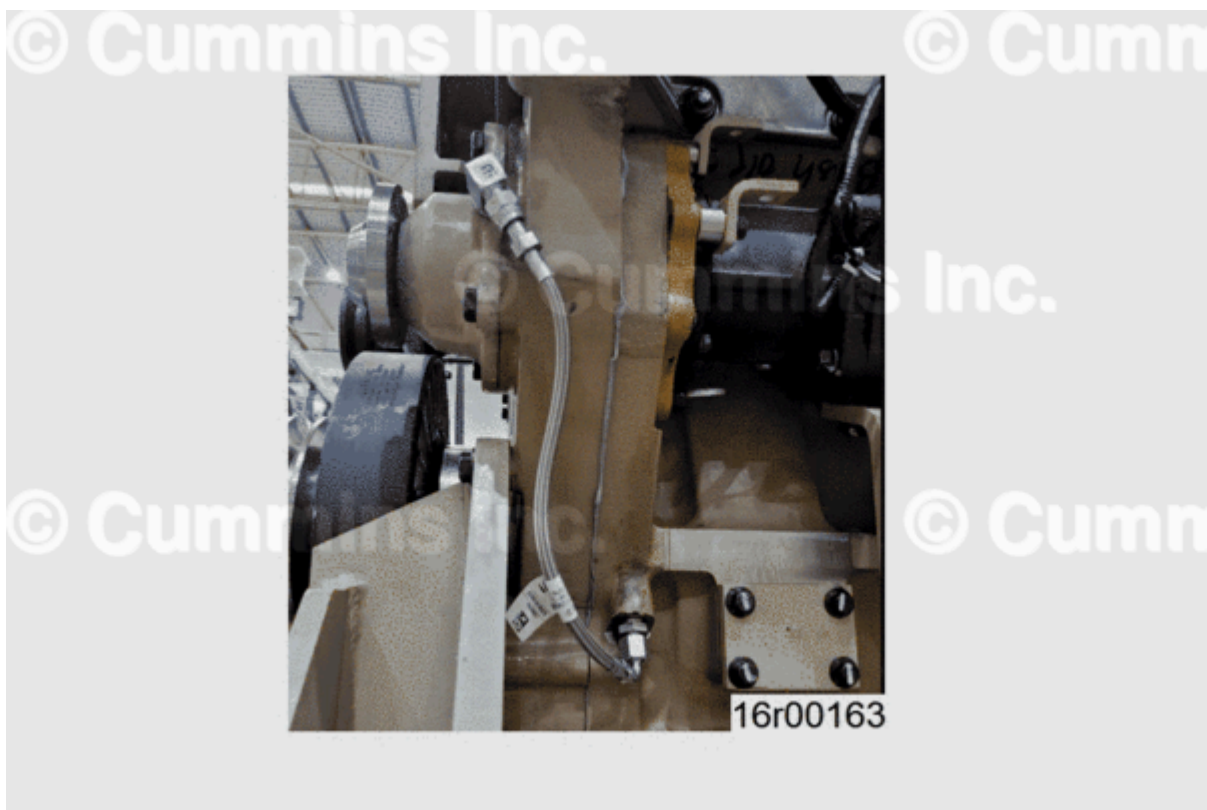


Figure 3, Elbow Installation Angle and Appropriate Clearance Example.

Service Parts

Service parts are available. See Table 2 for part numbers.

Table 2, Service Kits/Parts

Part Description	Existing Part Number	Obsolete	Superseded	New Part Number	Name	Comments	Usage
Drive, Fuel Pump / Support, Accessory Drive	5633425	No	No	6574803	Kit, Service	New Accessory Drive Kit WITH External Oil Feed Line	QSK23 CM2350 K127
Drive, Fuel Pump / Support, Accessory Drive	5633425	No	No	6574804	Kit, Service	New Accessory Drive Kit WITHOUT the External Oil Feed Line	QSK23 CM500 HPI, QSK23 CM2250 HPI K109

Table 2, Service Kits/Parts							
Part Description	Existing Part Number	Obsolete	Superseded	New Part Number	Name	Comments	Usage
Drive, Fuel Pump / Support, Accessory Drive	5633425	No	No	6574805	Kit, Service	New Accessory Drive Kit WITH External Oil Feed Line	QSK23 CM500 HPI, QSK23 CM2250 HPI K109

Part Compatibility

- The new accessory drive support kits are compatible with both the 75A and 140A alternator configurations.
- Accessory drive support kits, Part Number 6574803 and 6574805, are **not** to be used on engines equipped with air compressor and accessory drive option DF5730 because of lubricating oil supply needs of the air compressor.
- See Technical Service Bulletin, Loose or Worn Accessory Drive Assembly and Damage Mitigation Recommendations, TSB230177 if the new accessory drive with direct feed external oil supply is **not** able to be used.

Production Status

Implemented for production. See Table 3.

Table 3, Production Information			
Product	*Engine Serial Number (ESN) First	Build Date	Engine Plant
QSK23 CM500 FCO	85020888	20 September 24	HHS - CTIL HIGH-HORSEPOWER (SEZ) (Phaltan Engine Plant)
*ESNs with DF5730 (Air Compressor accommodation option) after ESN first are built with the improved accessory drive but without direct oil feed external line.			

Document History

Date	Details
2025-3-25	Module Created

Date	Details
2025-6-4	Added Step 7.

Last Modified: 04-Jun-2025
